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Sniping in soft-close online auctions: empirical evidence from overstock

Wen Cao¹ · Qinyang Sha² · Zhiyong Yao¹ · Dingwei Gu³ · Xiang Shao¹

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Abstract

The existing studies suggest that sniping is an equilibrium strategy in hard-close online auctions, but not in soft-close ones. In this paper, we use a unique, large-scale data set from soft-close Overstock and hard-close eBay to document sniping phenomena under the two different closing rules. Estimation results show that sniping is prominent on both websites, but they are prevalent at different times. On eBay, sniping occurs right before the auction close, while on Overstock sniping happens predominantly in a short window of time before the triggering period, during which any additional high bid automatically extends the online auction. Furthermore, the revenue effect of sniping is significantly stronger on Overstock than on eBay.

Keywords Sniping · Hard-close auctions · Soft-close auctions · Revenue effect

✉ Zhiyong Yao
 yzy@fudan.edu.cn

Wen Cao
 wenapply@gmail.com

Qinyang Sha
 qinyangsha@gmail.com

Dingwei Gu
 Dingwei.gu@gmail.com

Xiang Shao
 xshao@fudan.edu.cn

¹ School of Management, Fudan University, 670 Guoshun Road, Shanghai 200433, China

² Megaputer Intelligence, Bloomington, IN, USA

³ School of Economics, Fudan University, Shanghai 200433, China

1 Introduction

Sniping is a last-minute auction bidding practice wherein many bids are submitted close to or at the very end of an auction. Online auction platforms like eBay have made hard-close auctions a key feature of their business model, but other auction types do exist. For example, Overstock uses soft-close auctions that extend the time of an auction if a bid is submitted within the last 10 min. Although sniping in hard-close auctions is to be expected, due to its obvious strategic advantages, our data surprisingly indicate that a similar amount of sniping takes place in soft-close Overstock auctions.¹ This finding contradicts the existing literature (Malaga et al. 2010; Ockenfels and Roth 2006; and many others) that claims sniping to be an ineffective and suboptimal strategy for soft-close auctions.

To illustrate our finding, Fig. 1 plots the relative timings of about 670,000 bids on items listed for auction by large jewelry retailer Neimans Jewelry on hard-close eBay and soft-close Overstock auctions between November 2006 and October 2007. The distributions show that, on both websites, about 20% of the bids were placed during the last 1% of each auction's duration. Our current study focuses on this neglected phenomenon of sniping in soft-close auctions.

This paper examines this discrepancy by analyzing a unique, large-scale data set derived from real auction data on eBay and Overstock to observe sniping phenomena under two different auction closing rules. Our results contribute to the literature on online auctions in the following ways. First, sniping proves to be prevalent at different times for different auction types. In eBay hard-close auctions, sniping occurs right before auction close. However, in Overstock soft-close auctions, sniping behavior occurs in the minute before the trigger threshold, which occurs at the 10-min mark before auction close. Second, we examine the revenue effects of sniping on these two platforms and conclude that Overstock has a much stronger revenue effect from sniping than eBay does. Understanding these subtle, but critical, differences between hard-close and soft-close auctions is essential for revenue managers who make strategic sales decisions.

We empirically document that sniping may be more prevalent and more profitable in soft-close auctions than in hard-close auctions. Two possible reasons may contribute to this phenomenon. On the one hand, sniping displays the characteristics of tacit collusion during the very last moment before triggering the extended period. The explicit last moment offers time for coordination of tacit collusion, in which all bidders submit bids that are much lower than their true valuations. The potential bidding wars in the extended period and other identical goods available at the same auction website help to sustain this equilibrium. On the other hand, bidders with lower opportunity cost are more likely to snipe in soft-close auctions. In hard-close auctions, it is almost costless to snipe, which can even be achieved by using software. However, in soft-close auctions, bidders must exert significantly higher monitoring cost to place winning snipe bids, as they not only have to snipe at the very last moment before triggering the extended

¹ Overstock extends the auction for an additional 10 min when there is a new high bidder within 10 min of the scheduled auction closing time. Such a bid automatically extends the auction for another 10 min, and so on, until no more bids occur within the last 10 min, at which time the soft-close auction is closed.

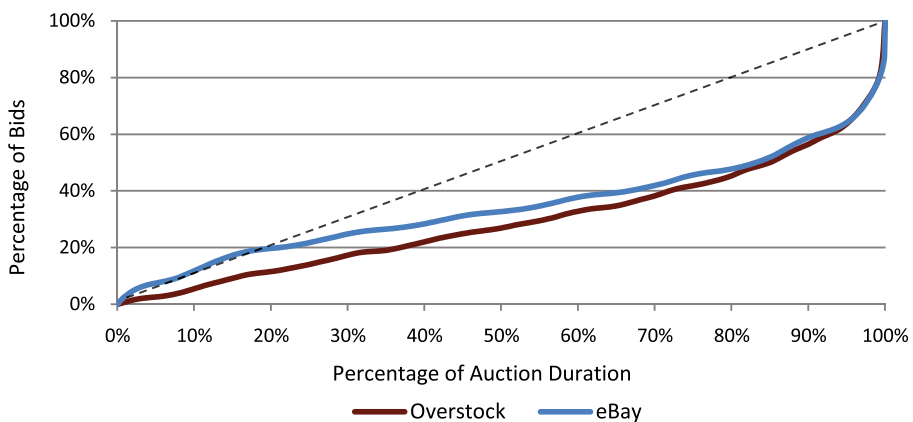


Fig. 1 Relative timing of bids in online auctions

period, but also need to monitor whether new bids are placed as the ending time is extended.²

Our findings have strong managerial implications. We show that soft-close auctions have no inherent advantage in mitigating sniping, as the existing literature suggests. Therefore, the efforts of online venues that apply soft-close auctions to mitigate sniping might be in vain. In addition, our analysis of the revenue effect proves that sniping in soft-close Overstock auctions does more damage to the seller compared with hard-close eBay auctions. That is, auction venues that use soft-close auctions to mitigate sniping and thus improve the sellers' profit may be working in the wrong direction. Sniping in soft-close auctions should pique the interest of any marketer, salesperson, or researcher in the online auction industry.³

2 Related literature

Studies have examined bidder behavior in online auctions since almost the advent of online auctions (Park and Bradlow 2005). General descriptions of such phenomena are sniping (Roth and Ockenfels 2002), forward-looking (Zeithammer 2006), herding (Simonsohn and Ariely 2008), and overbidding (Feng et al. 2016).

Various rationales have sought to explain sniping behavior in hard-close auctions. For example, sniping was proved to be an equilibrium strategy under pure common value auctions (Bajari and Hortaçsu 2003) and interdependent values auctions (Schindler 2003). In some scenarios, the optimal solution prompts bidders to snipe to prevent revealing private information to a rival (Rasmusen 2006) or uninformed bidder (Hossain 2008) or to avoid bidding wars (Ockenfels and Roth 2006). However, none of the available extant theoretical literature on this subject predicts the existence of sniping in soft-close auctions. Most of the available scholarship on sniping precludes even the possibility of this type of result. Thus, our empirical results establish a new precedent on which future theoretical research must be pursued.

² We thank an anonymous referee for offering this explanation.

³ Interestingly, Overstock.com shut down its auction channel in 2011.

The current empirical literature examines a few of these bidding relationships in depth. First, Roth and Ockenfels (2002) and many others find that more sniping occurs in hard-close eBay auctions than in soft-close Amazon auctions. Second, the experience of an auction bidder affects his or her tendency to snipe. Wilcox (2000) finds that the probability of sniping is positively correlated with bidder feedback scores and the likelihood of the presence of interdependent values. Livingston (2010) describes the tendency of new eBay users to bid earlier than users with even a hint of experience. Our study supports these conclusions by finding that bidder experience leads to more sniping in both auction types, although this effect is insignificant on Overstock.

Third, previous literature on the revenue effect of different closing rules for online auctions offers mixed results. Houser and Wooders (2005) find that soft-close auctions yield higher seller revenues than hard-close ones. By contrast, Brown and Morgan (2009) use robust statistical tests to show that eBay has higher auction revenues than Yahoo. Glover and Raviv (2012) demonstrate that sales using the soft-close rule yielded selling prices that were 13 to 20% higher than the hard-close format. Gray and Reiley (2013) find evidence of lower prices in sniped auctions, but the minimal price difference of 2.54% is not statistically significant. Our empirical study confirms the positive revenue effect for soft-close auctions over hard-close ones. We also provide statistically significant evidence that the revenue effect of sniping on Overstock is as strong as, if not stronger than, that found on eBay.

3 Data and methodology

3.1 Data

Established in 2004, Overstock is a second-tier American auction website that used the soft-close rule. Except for this closing rule, most of its standard online auction features are the same as eBay. Our study examines auctions of jewelry items sold by Neimans Jewelry, a wholesale liquidator. We chose Neimans because it provides the perfect platform for assessing the impact of different auction closing rules for two reasons: Neimans had a large presence on both websites with similar inventories during our sample period; and it uses the same listing page design, description, and item catalog numbers for the same products on both sites.

We used Perl spider scripts to capture data from eBay and Overstock. Our first data set contained 24,339 items offered by Neimans Jewelry on eBay between November 1, 2006 and October 31, 2007. After cleaning flawed listings,⁴ our data set contained 17,608 usable observations. Our second data set is based on 29,144 Neimans listings on Overstock during the same time period. Of the Overstock auctions, 21.25% did not result in a sale, which left 22,386 usable observations in the sample. Overall, eBay constitutes about 45% of our auction data.

In our Overstock data, about 30% of the auctions in the sample were extended. Among these, about two-thirds had only one or two extensions, and about 6% extended for more than an hour. The median number of extensions among the extended auctions was two, and the longest extension was for 2 h and 40 min, with 22 rounds of extensions.

⁴ These listings could have had issues because they were canceled, ended prematurely by Buy-It-Now purchases, ended without any bid whatsoever, or simply contained technical errors.

For each auction in the final sample, our data include demographic information about the number of bids, bid time, number of bidders, each bidder's feedback score, and the amount of each bid. We also have auction-specific variables, including starting and closing times. Item-specific information contains the catalog number and the reference price, which is a replacement value that Neimans estimates. Around 99.65% of the auctions in the sample started at \$0.99. A small minority of the auctions, only 52 eBay and 365 Overstock listings, started at prices greater than \$100. When a product was offered repeatedly, the starting price usually stayed at \$0.99, but sometimes it would jump to a level in the vicinity of recent winning prices.

We follow Livingston (2010) and code two dummy variables for bidders' experience based on their respective feedback scores. As feedback scores are not comparable across the systems, we code them differently. For Overstock bidders, if the user's feedback score was at least 2 (10), or if the number of transactions participated in was at least 5 (25), then the low-level (high-level) experience dummy takes the value 1. For eBay bidders, if the user's feedback score was at least 10 (50), then the low-level (high-level) experience dummy takes the value 1. Table 1 presents the descriptive statistics of the key variables.

3.2 Methodological issues

Before delving into the data analysis, several methodological issues found in the empirical literature on late bidding and sniping are worth noting.

3.2.1 Coding of sniping in soft-close auctions

Although sniping in hard-close auctions is well-defined, sniping in soft-close auctions is difficult to determine, as auctions often are extended beyond the pre-announced closing time. In this paper, bidding time is the time left relative to the scheduled ending time seen by the bidder at the time of bidding. The optimal time for sniping behavior is in the minute before the reported and publicly available threshold for extension in a soft-close auction. According to the rules of Overstock soft-close auctions, as described in footnote 1, sniping behavior occurs 10 min prior to scheduled close. Therefore, our study singles out the 1 min before this initial 10-min trigger, defined as the 10- to 11-min window, for a separate regression.

3.2.2 Using the bidding time of first bids

The extant literature points to one overriding reason for sniping. Actors simply want to hide information. If bidding at the latest possible moment optimizes the concealment of information, all early bid strategies should be rejected. Any early-then-late bid combination reveals information to some extent. In this sense, last bids are not good proxies for testing the presence of sniping. Examining multiple bid indicators helps to alleviate this problem. Therefore, this study uses the timing of first bids instead, following Hossain (2008). In particular, we record two variables: (1) the timing of the first bids placed by the winning bidders at the auction level, and (2) each bidder's first bid at the bidder level. Snipers are defined as those auction participants who place their first bid as close as possible to the auction closing time or as otherwise defined in section 3.2.1.

Table 1 Descriptive statistics

	Mean		Median		Std dev		Minimum		Maximum	
	Ostk	eBay	Ostk	eBay	Ostk	eBay	Ostk	eBay	Ostk	eBay
Winning price	178.9	144.8	134.45	114.63	249.24	132.35	1.24	4.25	19,380	3800.08
Minimum price	5.47	1.68	0.99	0.99	40.22	13.71	0.99	0.99	1799	479
Reference price	5361.79	5163.9	4989	4821.5	4136.05	3198.98	328	328	177,715	54,053
Featured	0.39	0.13	0.39	0.13	0.49	0.33	0	0	1	1
Number of bids	20.22	15.04	18	14	10.84	7.24	2	2	179	70
Number of bidders	7.27	7.84	7	8	2.64	2.62	2	2	21	22
Rush hour	0.98	0.63	1	1	0.001	0.004	0	0	1	1
Weekend/holiday	0.39	0.32	0	0	0.003	0.004	0	0	1	1
Shopping season	0.2	0.12	0	0	0.003	0.003	0	0	1	1
Winner FB (Lo)	0.64	0.78	1	1	0.48	0.42	0	0	1	1
Winner FB (Hi)	0.33	0.54	0	1	0.47	0.5	0	0	1	1
Last bidder FB (Lo)	0.63	0.78	1	1	0.48	0.41	0	0	1	1
Last bidder FB (Hi)	0.35	0.55	0	1	0.48	0.5	0	0	1	1

Rush hour is defined as the busy evening hours between 16:00 and 21:00 U.S. Pacific Time. Weekend indicates the period from 16:00 on Friday through Sunday U.S. Pacific Time. Holidays include New Year's Day, Memorial Day, Independence Day, Labor Day, Thanksgiving, and Christmas. Shopping season is between November 1 and December 23. *FB*, first bid

3.2.3 Interchangeable use of late bidding and sniping

Late bidding occurs for strategic considerations, such as the desire to conceal information, and nonstrategic considerations, such as late arrival. The interchangeable use of late bidding and sniping blurs the strategic reasoning for sniping. Thus, tests using long intervals may fail to address the sniping behavior that is our focus. More importantly, the difference between late bidding and sniping may be especially important for studies comparing the two closing rules. Because the trigger periods on soft-close websites are typically only 2 to 10 min, it is clearly important to distinguish between the two practices in empirical studies.

Our study sets 1 min as the sniping threshold. Bids placed within the last minute of the theoretically optimal sniping time are defined as “sniping” bids. Bids placed late yet prior to this 1-min cushion are defined as non-sniping “late” bids. In the next section, we further demonstrate, using our empirical results, that the use of 1 min as the sniping threshold is not an arbitrary choice. There are statistically significant differences in the expected bidding probabilities for the period within the 1-min window and the periods outside it.

3.2.4 Using non-cumulative intervals

In this study, we use non-cumulative measures when coding late bidding binary variables. To ensure robustness, empirical studies of late bidding typically use a series of thresholds, such as 1, 5, and 10 min from auction close, and binary variables are then coded according to such thresholds, respectively. Furthermore, this coding is done cumulatively. In particular, for thresholds t_1 and t_2 , $t_1 < t_2$, cumulative counts in the intervals of $[T - t_1, T]$ and $[T - t_2, T]$ are used. However, the problem with this coding is that a large effect in $[T - t_1, T]$ may overshadow a smaller but significant effect in $[T - t_2, T - t_1]$, which makes the robustness check not so robust.

To remedy this issue, we code the binary late-bidding dependent variables using non-cumulative intervals, such as (0, 10 s), (10 s, 30 s), (30 s, 1 min), (1 min, 5 min), and six more intervals up to 3 h. We then run regressions on these different intervals to describe late-bidding patterns during different intervals and check for robustness.

4 Analysis

4.1 Regression

We use probit regressions to examine the determinants of sniping and non-sniping “late” bidding with the data, focusing on the effects of the different closing rules.

As discussed in the previous section, we use the winning bidder’s first bid (WBFB) to construct a series of binary response variables with the non-cumulative interval as our dependent variable.⁵ Following existing studies and anecdotal information, we use 10 s, 30 s, and 1 min as the thresholds to investigate sniping. In contrast, we use 10 min, 30 min, and 60 min as the thresholds for non-sniping “late” bidding, together with

⁵ We also code the first bid of the last bidder or all bidders as a dependent variable to conduct robustness checks.

several other thresholds for comparison and reference purposes. We also single out the 10- to 11-min interval as the sniping window in soft-close auctions.

We estimate our baseline model with a probit model. Specifically, we have the following:

$$\Pr(T_i^p) = \Phi(\alpha + \text{EBAY}_i\beta_1 + X_i\gamma),$$

where $\Pr(T_i^p)$ is the probability that a WBFB in auction i is made within interval p . $\Phi(\cdot)$ is the cumulative normal distribution function. EBAY_i is our variable of interest, which is a dummy variable equal to 1 for hard-close auctions (eBay) and 0 for soft-close auctions (Overstock). X_i is a set of explanatory variables including the following (see also Table 2): *reference price* is an estimated replacement value, which is an important reference value when bidders submit their bidding; *featured* indicates whether the auction was featured on category and search results pages—sellers purchase this option to attract more bidders; *weekend/rush hour/shopping season* indicates whether the auction was scheduled to close during a weekend, evening, rush hour, holiday, or other seasonal shopping period—these variables may affect bidders' arrival process and thus the intensity of late bidding; and *had experience* indicates whether the bidder had initial experience on the platform at the time of bidding.⁶

Each auction constitutes one observation, and the time of the WBFB in that auction is checked against a series of time intervals to determine the values of the binary dependent variables. Table 2 reports the marginal effects obtained from these probit regressions with the predicted probabilities.⁷ Because the intervals are of different lengths, we rescaled the predicted probability within each interval to a per-second measure, as listed in the last two rows of the table. The standard error estimation could be biased as error terms may be correlated within products. Therefore, clustered standard errors are calculated at the product level (products with the same catalog number) to avoid this bias (Schmidheiny 2012).

For eBay, the predicted probability of an auction's WBFB increases monotonically in the last 3 h of the auction. There is a large jump in this probability during the last minute and a 14.93% chance that the WBFB is submitted in the last 10 s of the auction. Within the last minute of the auction, 28.19% of such bids are submitted. These findings are consistent with the existing literature claiming that sniping is prevalent in eBay auctions.

For Overstock, the predicted probability of an auction's WBFB increases monotonically in the last 3 h of the auction until it peaks in the 10- to 11-min sniping window. Then, during the last 10 min of the auction, the predicted probability drops slightly while remaining at a relatively high level. The peak in bidding activity during the 10- to 11-min window indicates that sniping is also prevalent in Overstock auctions.

Furthermore, the bidding within the last minute (relative to the virtual close time) is much weaker in Overstock auctions, with only a 1.4% chance for the WBFB to appear

⁶ We do not control the number of bidders, because no information about the current number of bidders or visitors is available for bidders at the time of making their bidding decision in real-life online auctions. Moreover, since bidders arrive to the auction at different times and never know the total realized number of bidders until the end of an online auction anyway, this variable is not reliable.

⁷ We have checked for the issue of multicollinearity. Please see the correlation matrix in the online appendix. Also it will be available upon request.

	(0, 10 s)	(10 s, 30 s)	(30 s, 1 min)	(1 min, 5 min)	(5 min, 10 min)	(10 min, 11 min)	(11 min, 30 min)	(30 min, 1 h)	(1 h, 2 h)	(2 h, 3 h)
Marginal effects										
eBay (vs. Overstock)	0.1445***	0.0826***	0.0368***	0.0154***	-0.0362***	-0.0124***	-0.0676***	-0.0274***	-0.0179***	-0.0121***
Reference price	5.32E-7***	6.22E-7***	1.65E-07	-7.31E-07	-2.36E-07	1.16E-07	-5.48E-08	-2.30E-07	-3.54E-07	3.92E-07
Featured	0.0023	-0.0011	0.0013	0.0025	6.14E-07	0.0006	-0.0074**	-0.0063**	-0.0058*	-0.0107***
Rush hour	0.0051***	0.0022*	0.0009	-0.0029	0.0046	-0.0009	-0.0092*	-0.0057	-0.0140***	-0.0089***
Weekend/holiday	-0.001	-0.0019	0.0007	-0.0070**	-0.0070***	0.0004	-0.0076***	-0.0037	-0.0048*	-0.0001
Shopping season	-0.0019	0.0009	-0.0028	-0.0146***	-0.0177***	-0.0022*	-0.0288***	-0.0104***	-0.0108***	-0.0043
Had experience	0.0143***	0.0063***	0.0031**	-0.0034	0.0090***	0.0001	-0.0013	-0.0057**	-0.0113***	-0.0054***
Predicted prob. (%)	14.93	8.83	4.43	7.83	2.96	0.47	5.21	4.19	5.42	3.67
Overstock	0.19	0.45	0.76	6.13	6.48	1.68	11.26	6.55	6.61	4.35
eBay (cumulative)	14.93	23.76	28.19	36.02	38.98	39.45	44.66	48.85	54.27	57.94
Overstock (cumulative)	0.19	0.64	1.4	7.53	14.01	15.69	26.95	33.5	40.11	44.46
eBay (per sec)	1.493	0.4415	0.1477	0.0326	0.0099	0.0078	0.0046	0.0023	0.0015	0.001
Overstock (per sec)	0.019	0.0225	0.0253	0.0255	0.0216	0.028	0.0099	0.0036	0.0018	0.0012

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in such sniping windows. This result may be due to late arrivals. There is statistically significantly more last-minute bidding on eBay than on Overstock. However, the marginal effects show that the large scale of late bidding on Overstock is statistically significant relative to eBay after the last 5-min interval. For all intervals between 5 min and 3 h from auction close, the sign switches and is still statistically significant. In other words, these intervals show a statistically significant amount of additional non-sniping “late” bidding on Overstock versus eBay.

The predicted probabilities also show a clear pattern of non-sniping “late” bidding on both websites. Not only does the increasing trend persist throughout the last 12 h of the auction, but some of the growth rates of these probabilities in the last hour are much higher than those in the earlier periods. Excluding the respective sniping windows, about 20.65% of the WBFs on eBay and 31.8% of those on Overstock are predicted to appear during the last hour, and 16.47% and 25.27%, respectively, in the last 30 min.

To summarize, our results show that the sniping time window is different for auction sites with different closing rules. Although non-sniping “late” bidding is prevalent under both closing rules, the data indicate the prominence of sniping in soft-close auctions.

The results also show that the other independent variables have mixed effects on sniping. For example, *reference price* seems to play a significant role in the sniping stage. Surprisingly, leisure time variables (e.g., *weekend* and *shopping season*) are significantly negatively related to late bidding probabilities in the final non-sniping intervals. *Experience* generally leads to more late bidding by the winning bidders. Interestingly, experience is also a significant indicator of sniping on eBay, but it does not demonstrate the same predictive value for sniping on Overstock.

As a robustness check, we conducted similar probit regressions using combinations of the following five dimensions of the data: (1) last bidder instead of winning bidder, (2) bidders with different levels of experience, (3) bidder level instead of auction level, (4) auctions matched at different levels, and (5) auctions’ last bid rather than first bid. The estimation and prediction results are qualitatively similar to the results we obtain in the baseline model for all five cases.⁸

4.2 Revenue effect

Theory suggests that sniping leads to lower seller revenue. We use an ordinary least squares model to check for the revenue effects of sniping bids under different closing rules. The dependent variable is the log of the winning bid price. Previous studies (e.g., Chan et al. 2007; Yao and Mela 2008) have proven that not only item characteristics and bidder behavior affect bidder demand, but type of market and auction also affect bidder demand. In addition to the variables in our baseline model, we add a set of dummy variables for auction length and variables for the intensity of bidding competition in the auction (the number of bidders within the time interval or last 30 min, number of multi-bid bidders who have submitted multiple bids during the auction, median timing of first bids, and maximal number of bids per bidder). Moreover, an interaction term between the website (eBay = 1) and bid timing (sniping = 1) dummies is included to show the difference in sniping

⁸ These regressions are omitted to save space, but they are available upon request.

effects between these two websites. In this analysis, the sniping dummy for soft-close auctions is coded using the “double take” rule, which means that winning bids are placed within the sniping window before the initial trigger period or within the new extended trigger period.

Table 3 displays the regression results. For non-sniping bids, the average winning price is lower on eBay than on Overstock. On both websites, a sniping winning bid for the most part is associated with a revenue loss compared with a non-sniping winning bid. For example, winning bids submitted within the 10-s window on Overstock (eBay = 0, sniping = 1) yield a decrease of 8.48% in seller revenue. Similar sniping on eBay (eBay = 1, sniping = 1) gives a much smaller decline of 2.21% ($= -0.0848 + 0.0627$) in seller revenue. Interestingly, as shown at the bottom of Table 3, the revenue effect of a sniping bid on Overstock is as strong as, if not stronger than, that on eBay.

Table 3 Ordinary least squares estimates of revenue effects

Dependent: log of winning price	(0, 10 s)	(10 s, 30 s)	(30 s, 1 min)	(0, 1 min)
Constant	−1.7531***	−1.7512***	−1.7504***	−1.7513***
eBay (vs. Overstock)	−0.0602***	−0.0614***	−0.0645***	−0.0614***
Sniping	−0.0848***	−0.0106	−0.0515***	−0.0454***
eBay x sniping	0.0627**	0.0001	0.0903***	0.0357**
Log of reference price	0.7032***	0.7030***	0.7031***	0.7030***
Log of starting price	0.1696***	0.1696***	0.1695***	0.1697***
Featured	0.2033***	0.2032***	0.2032***	0.2034***
Rush hour	0.0211*	0.0210**	0.0208**	0.0211*
Weekend/holiday	0.0112	0.0112	0.0111	0.0112
Shopping season	0.1199***	0.1198***	0.1198***	0.1198***
Had experience	−0.0558***	−0.0562***	−0.0567***	−0.0561***
Number of bidders	0.0316***	0.0314***	0.0312***	0.0316***
Number of bids in the last 30 mins	0.0058***	0.0057***	0.0057***	0.0058***
Number of multi-bid bidders	0.0458***	0.0462***	0.0462***	0.0459***
Median timing of first bids	4.31E-5***	4.32E-5***	4.33E-5***	4.31E-5***
Max number of bids per bidder	0.0130***	0.0130***	0.0130***	0.0129***
1-day auction	−0.1947***	−0.1906***	−0.1903***	−0.1924***
3-day auction	−0.1876***	−0.1868***	−0.1893***	−0.1866***
5-day auction	0.4140***	0.4136***	0.4133***	0.4132***
10-day auction	−0.0994***	−0.0993***	−0.0989***	−0.0992***
Effects of sniping on eBay	−0.0221***	−0.0104***	0.0388***	−0.0097***
Effects of sniping on Overstock	−0.0848***	−0.0106	−0.0515***	−0.0454***

Robust standard errors are calculated by clustering at the product level. The 7-day duration is omitted

*, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5 Conclusion

The existing literature indicates that sniping is an equilibrium strategy in hard-close online auctions, but not in soft-close ones. This paper uses a unique, large-scale data set from soft-close Overstock and hard-close eBay auctions to document sniping phenomena under these two different closing rules. Despite the different auction closing rules, our estimation results show that sniping is prominent on both websites, at different times relative to the auction close. Furthermore, the revenue effect of sniping is stronger on Overstock than it is on eBay. As online auction platforms become more critical to the global economy, even the slightest edge in understanding online auction strategy could mean billions of dollars in economic gains. Thus, the need for more theoretical work is urgent.

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